💳 🛡️ CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

🌟 CODSOFT DATA SCIENCE INTERNSHIP – TASK 5

📌 OBJECTIVE 🎯

The objective of this project is to build a machine learning model that detects fraudulent credit card transactions using classification techniques.

The model helps to identify whether a transaction is:

✔ Normal (0)

❌ Fraud (1)

📁 PROJECT DESCRIPTION

This project uses a credit card dataset containing transaction details and classified labels.

Dataset Features:

⏱ Time

💰 Amount

🔢 V1 to V28 (PCA transformed features)

🏷️Class (Target Variable)

🛠️TOOLS USED

Python 🐍

Pandas

NumPy

Matplotlib 📊

Seaborn

Scikit-learn 🤖

VS Code 💻

📊 MODEL USED

Logistic Regression (Classification Algorithm)

📈 MODEL PERFORMANCE

Accuracy: 99.91%

Precision (Fraud): 0.86

Recall (Fraud): 0.61

F1-Score (Fraud): 0.72

📌 DATASET INFORMATION

Total Records: 284,807

Normal Transactions: 284,315

Fraud Transactions: 492

🎯 SAMPLE OUTPUT

✔ Dataset loaded successfully

✔ Model trained using Logistic Regression

✔ Classification report generated

✔ Confusion matrix plotted

📊 CONFUSION MATRIX

Shows the performance of the model:

True Positives (Fraud detected correctly)

True Negatives (Normal detected correctly)

False Positives

False Negatives

🧠KEY CONCEPTS USED

✔ Data preprocessing

✔ Feature scaling

✔ Train-test split

✔ Classification model

✔ Evaluation metrics

✔ Confusion matrix

💡 KEY LEARNINGS

✔ Understanding fraud detection systems

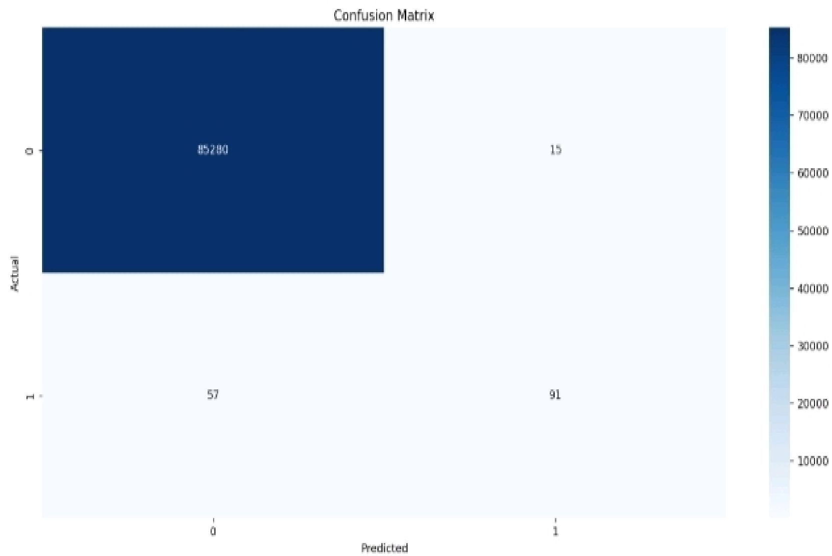
✔ Working with imbalanced datasets

✔ Importance of recall in fraud detection

✔ Real-world machine learning application

📸Output

Confusion Matrix Graph




```
Dataset Shape: (284807, 31)
   Time        V1        V2        V3        V4  ...       V26       V27       V28  Amount  Class
0   0.0 -1.359807 -0.072781  2.536347  1.378155  ... -0.189115  0.133558 -0.021053  149.62      0
1   0.0  1.191857  0.266151  0.166480  0.448154  ...  0.125895 -0.008983  0.014724    2.69      0
2   1.0 -1.358354 -1.340163  1.773209  0.379780  ... -0.139097 -0.055353 -0.059752  378.66      0
3   1.0 -0.966272 -0.185226  1.792993 -0.863291  ... -0.221929  0.062723  0.061458  123.50      0
4   2.0 -1.158233  0.877737  1.548718  0.403034  ...  0.502292  0.219422  0.215153   69.99      0

[5 rows x 31 columns]

Accuracy: 0.9991573329588147

Classification Report:
              precision    recall  f1-score   support
```

🏁 CONCLUSION

The model successfully detects fraudulent transactions with high accuracy. However, due to dataset imbalance, recall for fraud detection can be further improved using techniques like SMOTE or advanced ensemble models.

👩‍💻 AUTHOR

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